# Harry Potter Sentiment Analysis

*NRC, Bing, and AFINN Lexicon Analysis Across the Series*

## Analysis of Results

### NRC Sentiment Distribution

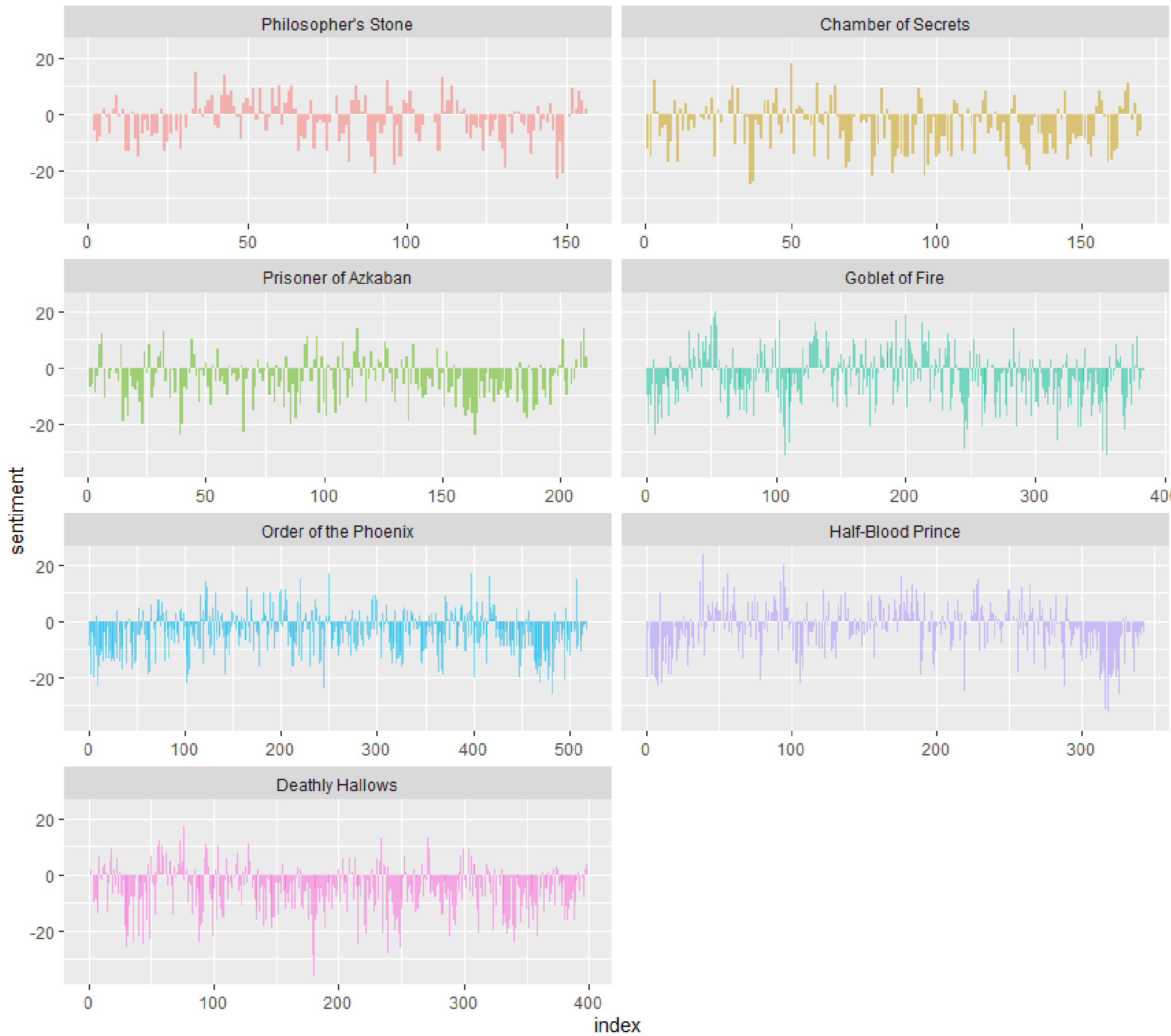
*Overall sentiment category word counts across the full Harry Potter series.*

| sentiment | n |
|---|---|
| negative | 55,096 |
| positive | 37,767 |
| sadness | 34,883 |
| anger | 32,747 |
| trust | 23,160 |
| fear | 21,536 |
| anticipation | 20,629 |
| joy | 13,804 |
| disgust | 12,861 |
| surprise | 12,818 |

**Interpretation**: The overall sentiment distribution shows that negative words appear more frequently than positive words across the entire Harry Potter series. Negative emotions such as sadness, fear, and anger appear more frequently, which suggests the books plots contain substantial conflict and emotional tension. Positive emotions such as trust, anticipation, and joy also appear quite frequently, which suggests themes of hope and growth to balance out the tone of the series. It is a conflict-driven book series rather than being purely positive however.

## Sentiment Heatmap: Philosopher's Stone

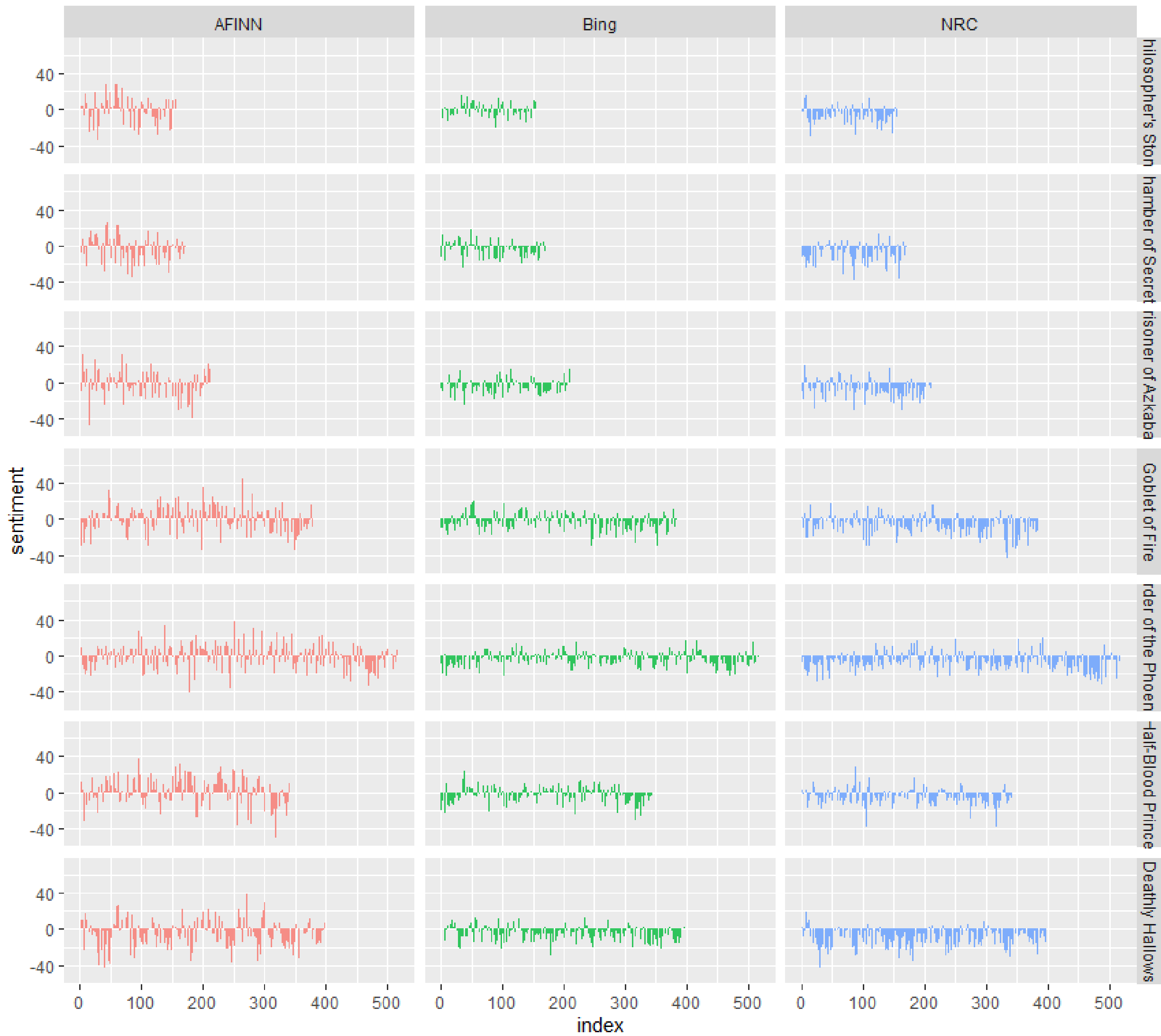
*Net AFINN sentiment score per chapter segment. Red = positive, blue = negative, white = neutral.*




**Interpretation**: The plots show net sentiment, positive to negative, per 500 word segment for each book. The sentiment fluctuates throughout each book however it is noticeably more negative as the series goes on, especially Deathly Hallows. Philosopher's Stone is the most neutral out of all the books.

## Top 10 Positive and Negative Words: Contribution to Sentiment

*Bing lexicon, word frequency contribution to overall sentiment score.*




**Interpretation**: All 3 sentiment methods reveal similar overall trends, which suggests that major emotional highs and lows are occurring in roughly the same places regardless of lexicon. AFINN produces the more larger peaks and valleys, since its assigns a weighted score. Bing and NCR have less extreme values due to classifying words as negative and positive.

## Most Common Sentiment Words: Bing Lexicon

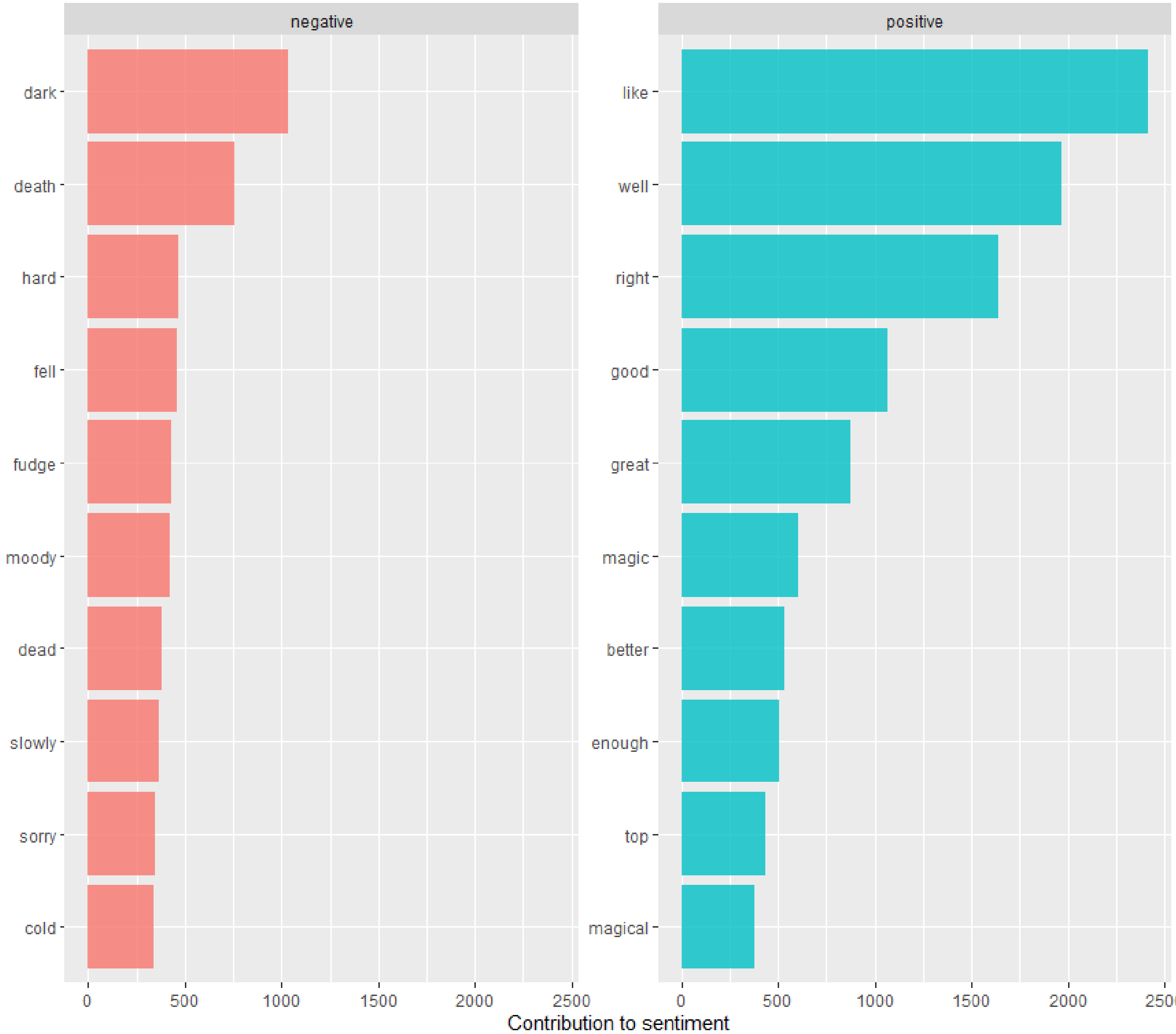
*Top 10 most frequently occurring positive and negative words across the series (top 10 of 3,313 unique words).*

| word | sentiment | n |
|---|---|---|
| **like** | positive | 2,416 |
| **well** | positive | 1,969 |
| **right** | positive | 1,643 |
| **good** | positive | 1,065 |
| **dark** | negative | 1,034 |
| **great** | positive | 877 |
| **death** | negative | 757 |
| **magic** | positive | 606 |
| **better** | positive | 533 |
| **enough** | positive | 509 |

**Interpretation**: The most frequent positive words are “like”, “well”, “right”, “good”, and “great”. These words suggest that much of the dialogue/narration includes expressions of encouragement and affirmation. The word “magic” appears among the frequent positive words which reinforces the central theme of the series (wizards). The most frequent negative words are “dark” and “death”. These words suggest that the narrative addresses danger, mortality, threatening situations, and symbolic uses of darkness throughout the series.

## Sentiment Comparison Across Lexicons

*AFINN, Bing, and NRC sentiment scores compared across all seven books.*




**Interpretation**: The bar chart shows the top 10 most frequent positive and negative words used in the series. Just like in the last interpretation, positive words such as “like”, “well”, “right”, “good”, “great”, and “magic” appear most frequently. The negative words that appear most frequently are “dark”, “death” (just like in the last interpretation), “hard”, “fell”, and “fudge”. This reinforces the idea that it is a conflict-driven book series rather than being purely positive.

## Sentiment by Chapter: Philosopher's Stone (AFINN)

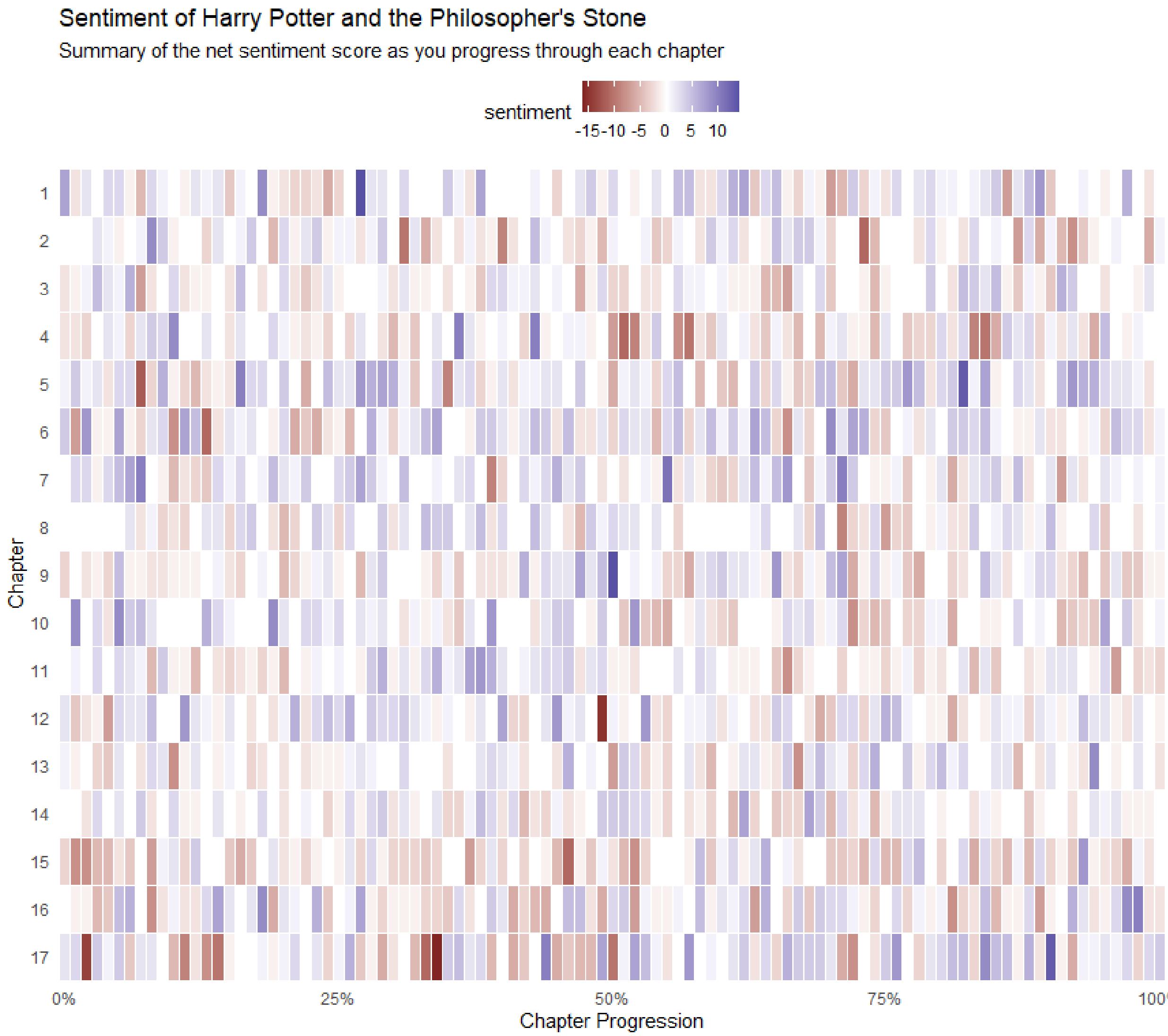
*Top 10 highest-sentiment segments in Harry Potter and the Philosopher's Stone, sorted by descending sentiment score.*

| chapter | index | sentiment |
| --- | --- | --- |
| 1 | 0.27 | 14 |
| 9 | 0.50 | 14 |
| 5 | 0.82 | 13 |
| 17 | 0.90 | 13 |
| 7 | 0.55 | 11 |
| 4 | 0.36 | 10 |
| 4 | 0.43 | 10 |
| 5 | 0.16 | 10 |
| 7 | 0.07 | 10 |
| 7 | 0.71 | 10 |

**Interpretation**: This table shows that the strongest positive sentiments in book 1 the Philosopher's Stone are chapters 1, 9, 5, and 17 (the last chapter). Chapter 9 has very strong positive moments because it is Harry's first flight lessons, also when he joins the Gryffindor Quidditch team, and when he goes on a midnight adventure. Chapter 7 appears quite frequently in the tibble, which is when the students get sorted into their houses by the sorting hat, which has its positive moments.

## Net Sentiment per 500-Word Segment by Book

*Bing lexicon, net sentiment (positive minus negative) across each book in the series.*




**Interpretation**: The heatmap shows the net sentiment per chapter in the Philosopher's Stone. Red means positive, blue means negative, and white means neutral. The sentiment clearly fluctuates per chapter, many chapters begin relatively neutral and then shift towards a stronger emotional tone near the middle and end. Later chapters display stronger contrasts between positive and negative, reflecting tension near the climax of the story. The final chapter shows more positive sentiment towards the beginning and middle, with more neutral sentiment towards the end, suggesting a resolution and victory in the final chapter. The patterns throughout the book clearly show a rising tension, climax, and eventful resolution.

## References

Boehmke, B. (n.d.). *harrypotter: An R package for the Harry Potter book series* [R package]. GitHub. https://github.com/bradleyboehmke/harrypotter

# Attachments

## R Outputs

```r
# Download package from Github
if (packageVersion("devtools") < 1.6) {
  install.packages("devtools")
}

devtools::install_github("bradleyboehmke/harrypotter")

# Load libraries
library(tidyverse)      # data manipulation & plotting
library(stringr)        # text cleaning and regular expressions
library(tidytext)       # provides additional text mining functions
library(harrypotter)    # provides the first seven novels of the Harry Potter series



# First two chapters
philosophers_stone[1:2]

# Sentiment data sets
sentiments

# to see the individual lexicons try
get_sentiments("afinn")
get_sentiments("bing")
get_sentiments("nrc")

# Basic Sentiment Analysis
titles <- c("Philosopher's Stone", "Chamber of Secrets", "Prisoner of Azkaban",
            "Goblet of Fire", "Order of the Phoenix", "Half-Blood Prince",
            "Deathly Hallows")

books <- list(philosophers_stone, chamber_of_secrets, prisoner_of_azkaban,
              goblet_of_fire, order_of_the_phoenix, half_blood_prince,
              deathly_hallows)

series <- tibble()

for(i in seq_along(titles)) {

  clean <- tibble(chapter = seq_along(books[[i]]),
                  text = books[[i]]) %>%
    unnest_tokens(word, text) %>%
    mutate(book = titles[i]) %>%
    select(book, everything())

  series <- rbind(series, clean)
}

# set factor to keep books in order of publication
series$book <- factor(series$book, levels = rev(titles))

series

# NRC sentiment data set
series %>%
  right_join(get_sentiments("nrc")) %>%
  filter(!is.na(sentiment)) %>%
  count(sentiment, sort = TRUE)
```

```r
# Index that breaks up each book by 500 words
# Assesses the positive vs negative sentiment of each word
# Counts how many positive and negative words there are for every two pages
# Calculate nt sentiment
# Plots data
series %>%
  group_by(book) %>%
  mutate(word_count = 1:n(),
         index = word_count %/% 500 + 1) %>%
  inner_join(get_sentiments("bing")) %>%
  count(book, index = index , sentiment) %>%
  ungroup() %>%
  spread(sentiment, n, fill = 0) %>%
  mutate(sentiment = positive - negative,
         book = factor(book, levels = titles)) %>%
  ggplot(aes(index, sentiment, fill = book)) +
  geom_bar(alpha = 0.5, stat = "identity", show.legend = FALSE) +
  facet_wrap(~ book, ncol = 2, scales = "free_x")

# Comparing sentiments
afinn <- series %>%
  group_by(book) %>%
  mutate(word_count = 1:n(),
         index = word_count %/% 500 + 1) %>%
  inner_join(get_sentiments("afinn")) %>%
  group_by(book, index) %>%
  summarise(sentiment = sum(value)) %>%  # had to change 'score' to 'value' from the tutorial for code to work
  mutate(method = "AFINN")

bing_and_nrc <- bind_rows(series %>%
                            group_by(book) %>%
                            mutate(word_count = 1:n(),
                                   index = word_count %/% 500 + 1) %>%
                            inner_join(get_sentiments("bing")) %>%
                            mutate(method = "Bing"),
                          series %>%
                            group_by(book) %>%
                            mutate(word_count = 1:n(),
                                   index = word_count %/% 500 + 1) %>%
                            inner_join(get_sentiments("nrc") %>%
                                         filter(sentiment %in% c("positive", "negative"))) %>%
                            mutate(method = "NRC")) %>%
  count(book, method, index = index , sentiment) %>%
  ungroup() %>%
  spread(sentiment, n, fill = 0) %>%
  mutate(sentiment = positive - negative) %>%
  select(book, index, method, sentiment)

# Bind the net sentiments together and plot
bind_rows(afinn,
          bing_and_nrc) %>%
  ungroup() %>%
  mutate(book = factor(book, levels = titles)) %>%
  ggplot(aes(index, sentiment, fill = method)) +
  geom_bar(alpha = 0.8, stat = "identity", show.legend = FALSE) +
  facet_grid(book ~ method)
```

```r
# Common Sentiment Words
bing_word_counts <- series %>%
  inner_join(get_sentiments("bing")) %>%
  count(word, sentiment, sort = TRUE) %>%
  ungroup()

bing_word_counts

# View top n words for each sentiment visually
bing_word_counts %>%
  group_by(sentiment) %>%
  top_n(10) %>%
  ggplot(aes(reorder(word, n), n, fill = sentiment)) +
  geom_bar(alpha = 0.8, stat = "identity", show.legend = FALSE) +
  facet_wrap(~sentiment, scales = "free_y") +
  labs(y = "Contribution to sentiment", x = NULL) +
  coord_flip()

# Sentiment analysis with larger units
tibble(text = philosophers_stone) %>%
  unnest_tokens(sentence, text, token = "sentences")

# breaking up philosophers_stone text by chapter and sentence
ps_sentences <- tibble(chapter = 1:length(philosophers_stone),
                       text = philosophers_stone) %>%
  unnest_tokens(sentence, text, token = "sentences")

# sentiments by chapter
book_sent <- ps_sentences %>%
  unnest_tokens(word, sentence) %>%          # had tokenize first, kept getting an error in 'group_by()'
  group_by(chapter) %>%
  mutate(sentence_num = row_number(),
    index = round(sentence_num / n(), 2)) %>%
  inner_join(get_sentiments("afinn")) %>%
  group_by(chapter, index) %>%
  summarise(sentiment = sum(value, na.rm = TRUE), .groups = "drop") %>% # again had to change 'score' to 'value'
  arrange(desc(sentiment))

book_sent

# visualization (heatmap) of sentiment as we progress through each chapter
ggplot(book_sent, aes(index, factor(chapter, levels = sort(unique(chapter), decreasing = TRUE)), fill = sentiment)) +
  geom_tile(color = "white") +
  scale_fill_gradient2() +
  scale_x_continuous(labels = scales::percent, expand = c(0, 0)) +
  scale_y_discrete(expand = c(0, 0)) +
  labs(x = "Chapter Progression", y = "Chapter") +
  ggtitle("Sentiment of Harry Potter and the Philosopher's Stone",
          subtitle = "Summary of the net sentiment score as you progress through each chapter") +
  theme_minimal() +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        legend.position = "top")
```

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